Course Outline (1/4)

- Week -01
  - Introduction to IR course, IR Problem and its Components, Basic IR model – Boolean Information Retrieval, Extended Boolean Model, Example of Commercial Systems – WestLaw <Ch: 1 >
- Week-02
  - Document Processing, Term vocabulary and positing lists, Stemming & Lemmatization, posting list processing via skip lists, Phrase Query, positional indexing, bi-word indexing, Combining different indexing techniques. <Ch: 2>

Course Outline (1/4)

- Week -03
  - Dictionary and Tolerant Retrieval, Search Structures for Dictionary, Wildcard queries, permuterm index, k-gram index, Spelling Correction, Edit Distance, Phonetic Correction <Ch: 3>
- Week-04
  - Index construction, single pass, in-memory, distributed indexing, dynamic indexing.
  - Heaps law, Zipf's law, dictionary indexing, fixed length and variable length coding, γ codes <Ch:4 and 5>

Course Outline (2/4)

- Week -05
  - Vector Space Model <Ch: 6 and Ch:7>
- Week-06
  - Midterm Exam
- Week-07
  - Evaluation of IR <Ch: 8>
- Week-08
  - Relevance Feedback <Ch: 9>

Course Outline (3/4)

- Week-9
  - Probabilistic Language Model for IR
  - Language Model for IR
- Week-10
  - Basic Web Search, Crawler and Indexing
  - Link Analysis <Ch: 19,20 and 21>
- Week-11
  - Midterm II
- Week-12
  - Text Classification/Clustering <Ch: 14,15, 16,17>

Course Outline (4/4)

- Week-13
  - Neural Information Retrieval
- Week-14
  - Learning Word Representation
- Week-15
  - RAG
- Week-16
  - Contrastive Learning

Grading Scheme

Assignments	20%
Quizzes	10%
Mid-Terms (2)	20%
Class Project	10%
Final	40%